Fundamental Introduction to Machine Learning

Week 5, Day 1

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Abstract

This session bridges the gap between raw clinical data and robust machine learning models. We begin with foundational Machine Learning concepts before diving into the critical early stages of the ML pipeline.

- **Foundations:** Definitions, types, and healthcare applications of ML.
- **Problem Formulation:** Defining the clinical objective and understanding the cost of prediction errors.
- **Exploratory Data Analysis (EDA):** Uncovering patterns, distributions, and relationships in the Heart Disease dataset.
- **Data Preparation:** Rigorous data partitioning and the strict prevention of data leakage using Scikit-Learn Pipelines.

Our ultimate goal is to build a reliable, clinically sound predictive model for Atherosclerotic Heart Disease (AHD).

Session Objectives

- Understand the core definitions, types, and applications of Machine Learning.
- Formulate the clinical problem of predicting Atherosclerotic Heart Disease (AHD).
- Evaluate the clinical implications of False Positives vs. False Negatives.
- Perform comprehensive Exploratory Data Analysis (EDA) to understand feature distributions and relationships.
- Implement a Stratified Train-Test Split to preserve class ratios.
- Master the concept of Data Leakage and prevent it using Scikit-Learn Pipelines.

Agenda

1. **Introduction to Machine Learning** (Definitions, Types, Applications)
2. **Problem Formulation & Clinical Context** (Objectives, Target Variable, Error Costs)
3. **Data Importation & Initial Inspection** (Loading data, handling missing values)
4. **Exploratory Data Analysis (EDA)** (Univariate, Bivariate, Multivariate analysis)
5. **Data Partitioning & Leakage Prevention** (Stratified splits, Pipelines, Cross-Validation)

Part 1: Introduction to Machine Learning

"From data to intelligent predictions."

What is Machine Learning?

Definition

Machine Learning (ML) is a subset of artificial intelligence that focuses on building systems that learn from data, identify patterns, and make decisions with minimal human intervention.

- **Arthur Samuel (1959):** "Field of study that gives computers the ability to learn without being explicitly programmed."
- **Tom Mitchell (1997):** A computer program learns from **Experience (E)** with respect to some **Task (T)** and measured by a **Performance Measure (P)**.
- **Core Idea:** Instead of writing explicit rules (e.g., 'if age ¿ 50 and HR ¡ 100 then disease'), we feed data to an algorithm to learn the rules itself.

Types of Machine Learning

- **Supervised Learning:** The algorithm learns from labeled training data (features + target).
  - *Example:* Predicting heart disease based on patient records.
- **Unsupervised Learning:** The algorithm finds hidden patterns in unlabeled data (features only).
  - *Example:* Grouping patients into distinct clinical phenotypes.
- **Reinforcement Learning:** An agent learns to make decisions by interacting with an environment and receiving rewards or penalties.
  - *Example:* Optimizing a dynamic treatment regimen over time.

Supervised Learning Deep Dive

Supervised learning is divided into two main categories based on the target variable:

Type	**Target Variable**	**Clinical Example**
Classification	Discrete / Categorical	Predicting if a patient has AHD (Yes/No)
Regression	Continuous / Numerical	Predicting a patient's exact systolic blood pressure

Our Focus

Today, we are tackling a **Binary Classification** problem: predicting the presence (1) or absence (0) of Atherosclerotic Heart Disease.

Unsupervised & Reinforcement Learning

Unsupervised Learning

- **Clustering:** K-Means, Hierarchical. (e.g., Discovering new patient subgroups).
- **Dimensionality Reduction:** PCA, t-SNE. (e.g., Compressing high-dimensional genomic data).

Reinforcement Learning (RL)

- Learns via trial and error using a **Reward System**.
- Used in robotics, autonomous vehicles, and game-playing AI (e.g., AlphaGo).
- In healthcare: Optimizing sequential treatment policies (e.g., sepsis treatment).

Applications of ML in Healthcare

- **Predictive Diagnostics:** Early warning systems for diseases like sepsis, heart failure, or diabetic retinopathy.
- **Medical Imaging:** Automated tumor detection in X-rays, MRIs, and CT scans using Deep Learning.
- **Drug Discovery:** Predicting molecular binding affinity and accelerating the drug development pipeline.
- **Personalized Medicine:** Tailoring treatment plans and drug dosages based on individual genetic and clinical profiles.
- **Operational Efficiency:** Predicting patient readmission rates and optimizing hospital bed allocation.

Part 2: Problem Formulation & Clinical Context

"Defining the clinical question and its real-world impact."

Problem Formulation & Clinical Context

- **Objective:** Predict Atherosclerotic Heart Disease (AHD) based on clinical and demographic indicators.
- **Why it matters:** Cardiovascular diseases are the leading cause of death globally. Early, accurate detection saves lives and reduces long-term healthcare costs.
- **The Dataset:** `Heart.csv` contains clinical data for 303 patients, including age, sex, chest pain type, blood pressure, cholesterol, and ECG results.

The ML Task

Map a set of input features X (clinical variables) to an output target y (disease presence) using a learned function $f(X) \approx y$.

Objective & Target Variable

- **Target Variable:** `AHD` (Atherosclerotic Heart Disease).
- **Type:** Binary Classification.
  - **0:** No Disease (Healthy)
  - **1:** Disease Present (AHD)
- **Features (Predictors):** 13 variables including:
  - **Demographic:** Age, Sex
  - **Symptoms:** ChestPain
  - **Vitals/Labs:** RestBP, Cholesterol, MaxHR, Oldpeak
  - **Diagnostics:** Ca (vessels colored), Thal (thalassemia)

Clinical Implications: Errors in Prediction

In medical diagnostics, not all errors are created equal.

False Negatives (Type II Error)

Missed cardiac risk. The model predicts "Healthy", but the patient actually has AHD. *Cost:* The patient is sent home untreated and may suffer a fatal heart attack. **Extremely high cost to human life.**

False Positives (Type I Error)

Unnecessary tests. The model predicts "Disease", but the patient is actually healthy. *Cost:* The patient undergoes unnecessary stress tests or angiograms. **High financial cost, psychological anxiety, but low risk to life.**

Why Recall/Sensitivity is Critical

Recall (Sensitivity)

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Measures the proportion of *actual* positive cases that were correctly identified.

- In healthcare, missing a sick patient (False Negative) is far more dangerous than a false alarm (False Positive).
- Therefore, optimizing for **high Recall** is often the primary clinical goal.
- We would rather flag 10 healthy patients for extra tests than miss 1 sick patient who needs immediate care.

Part 3: Data Importation & Initial Inspection

"Loading, cleaning, and understanding the raw data."

Data Importation and Handling Missing Values

Clinical datasets often contain missing values represented by symbols such as `?`, blanks, or `NA`. These must be converted to `NaN` before analysis.

```
import pandas as pd
import numpy as np
# Read dataset
df = pd.read_csv("Heart.csv", na_values=["?", "NA", "", " "])
# Count missing values
print(df.isnull().sum())
```

Key Points

- `na_values` converts predefined missing value indicators to `NaN`.
- Missing values can then be detected using `isnull()` or `isna()`.
- Proper handling of missing data prevents errors during statistical analysis and machine learning.

Initial Data Inspection

The first step after loading data is to understand its structure and quality.

Key Checks

- `df.shape`
  - Confirms dataset dimensions.
  - Heart dataset: 303 patients and 14 variables.
- `df.dtypes`
  - Verifies numeric variables (`Age`, `MaxHR`).
  - Verifies categorical variables (`ChestPain`, `Thal`).
- `df.isnull().sum()`
  - Identifies variables with missing values.
  - Guides the choice of imputation method.

Target Variable Distribution

Before building a classifier, examine the distribution of the outcome variable.

- Use:

```
df["AHD"].value_counts(normalize=True)
```

- The Heart dataset is approximately:
  - 55% Heart Disease
  - 45% No Heart Disease

Why Does This Matter?

A highly imbalanced dataset can produce misleadingly high accuracy. Even with a relatively balanced dataset, stratified train-test splitting should be used to preserve class proportions.

Part IV

Exploratory Data Analysis (EDA)

Discovering patterns, trends, anomalies, and relationships within the data.

What is Exploratory Data Analysis (EDA)?

EDA is the process of exploring and summarizing a dataset using descriptive statistics and visualization before predictive modeling.

Objectives of EDA

- Understand the distribution of variables.
- Detect missing values and outliers.
- Explore relationships among variables.
- Identify patterns associated with the target variable.
- Inform feature engineering and model selection.

"The greatest value of a picture is when it forces us to notice what we never expected to see."

— John Tukey

Univariate Analysis: Target Variable

Univariate analysis examines one variable at a time.

- Analyze the target variable independently.
- Visualize using a Count Plot or Pie Chart.
- Assess whether the classes are balanced.

Baseline Model

A simple classifier that always predicts the majority class would achieve an accuracy of approximately **55%**.

A useful machine learning model should substantially outperform this baseline.

Bivariate Analysis: Numerical Features

Compare numerical variables across the two outcome classes.

Visualization Techniques

- **Boxplots**: Compare median, spread (IQR), and outliers.
- **KDE Plots**: Display the probability density for each class.

Key Variables

- `Age`
- `MaxHR` (Maximum Heart Rate)
- `Oldpeak` (ST Depression)

Clinical Interpretation of Numerical Features

- **Age**: Patients with heart disease tend to be older.
- **MaxHR**: Diseased patients generally achieve lower maximum heart rates.
- **Oldpeak**: Higher values are associated with myocardial ischemia and heart disease.

Key Insight

The clear separation between the two classes suggests that these variables possess strong predictive value for machine learning models.

Bivariate Analysis: Categorical Features

Categorical variables are commonly compared using grouped or stacked bar charts.

- **ChestPain**: Asymptomatic patients show the highest disease prevalence.
- **Thal**: Reversible defects are strongly associated with heart disease.

Implication

Categorical predictors should be encoded (e.g., One-Hot Encoding) before model training.

Multivariate Analysis: Correlation Heatmap

- Measures Pearson correlation among numerical variables.
- Visualized using a color-coded heatmap.

Purpose

- Identify variables strongly associated with the target.
- Detect multicollinearity between predictors.
- Support feature selection.

Multivariate Analysis: Pairplots

- Pairplots visualize pairwise relationships between variables.
- Points are colored according to disease status.
- Focus on:
  - `Age`
  - `MaxHR`
  - `Oldpeak`
  - `Ca`

Insights

Pairplots reveal clusters, trends, and nonlinear relationships that may not appear in correlation matrices.

Part V

Data Partitioning and Leakage Prevention

Building reliable and unbiased machine learning models.

Train-Test Split

Evaluate models using data that were not seen during training.

```
from sklearn.model_selection import train_test_split

X = df.drop("AHD", axis=1)
y = df["AHD"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42
)
```

Dataset Split

- Training Set: 80%
- Test Set: 20%

Stratified Train-Test Split

Maintain the same class distribution in both datasets.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

Why Stratify?

Preserves the proportion of healthy and diseased patients in both the training and testing datasets, resulting in more reliable model evaluation.

Understanding Data Leakage

Definition

Data leakage occurs when information from the test set is unintentionally used during model training.

- Produces unrealistically high performance.
- Leads to poor real-world generalization.

Common Sources

- Scaling before splitting
- Global imputation
- Target-based feature engineering

Preventing Data Leakage with Pipelines

Fit preprocessing only on the training data.

```python
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler

pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

X_train = pipeline.fit_transform(X_train)
X_test = pipeline.transform(X_test)
```

Cross-Validation Without Leakage

```python
from sklearn.pipeline import Pipeline
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression

model = Pipeline([
    ("preprocessor", pipeline),
    ("classifier", LogisticRegression(max_iter=1000))
])

scores = cross_val_score(model, X_train, y_train, cv=5)
```

Best Practice

Every preprocessing step (imputation, scaling, encoding, feature selection) should be performed inside the Pipeline so that it is repeated independently within each cross-validation fold.

Summary & Best Practices

1. **Define the Problem:** Always understand the clinical context and the asymmetric cost of False Negatives vs. False Positives.
2. **Explore Thoroughly:** Use EDA to guide feature engineering and understand data distributions before modeling.
3. **Split First:** Always partition your data into train and test sets *before* any preprocessing.
4. **Prevent Leakage:** Use Scikit-Learn `Pipelines` to ensure preprocessing is fitted strictly on training data.
5. **Choose the Right Metric:** In healthcare, prioritize Recall/Sensitivity over raw Accuracy to minimize missed diagnoses.

Q&A and Thank You!

Questions?

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